

earlier with my example of survival rates, are not psychologically so. Worse, even professionals are fooled and base their decisions on their perceptual errors. Our research shows that the way a risk is framed sharply influences people's understanding of it. If you say that, on average, investors will lose all their money every thirty years, they are more likely to invest than if you tell them they have a 3.3 percent chance of losing a certain amount every year.

The same is true of airplane rides. We have asked experimental participants: "You are on vacation in a foreign country and are considering flying a local airline to see a special island. Safety statistics show that, if you fly once a year, there will be on average one crash every thousand years on this airline. If you don't take the trip, it is unlikely you'll visit this part of the world again. Would you take the flight?" All the respondents said they would. But when we changed the second sentence so it read, "Safety statistics show that, on average, one in a thousand flights on this airline have crashed," only 70 percent said they would take the flight. In both cases, the chance of a crash is 1 in 1,000; the latter formulation simply sounds more risky.

THE PROBLEM OF INDUCTION AND CAUSATION IN THE COMPLEX DOMAIN

What Is Complexity? I will simplify here with a functional definition of complexity—among many more complete ones. A complex domain is characterized by the following: there is a great degree of interdependence between its elements, both temporal (a variable depends on its past changes), horizontal (variables depend on one another), and diagonal (variable A depends on the past history of variable B). As a result of this interdependence, mechanisms are subjected to positive, reinforcing feedback loops, which cause "fat tails." That is, they prevent the working of the Central Limit Theorem that, as we saw in [Chapter 15](#), establishes Mediocristan thin tails under summation and aggregation of elements and causes "convergence to the Gaussian." In lay terms, moves are exacerbated over time instead of being dampened by counterbalancing forces. Finally, we have nonlinearities that accentuate the fat tails.

So, complexity implies Extremistan. (The opposite is not necessarily true.)

As a researcher, I have only focused on the Extremistan element of complexity theory, ignoring the other elements except as a backup for my considerations of unpredictability. But complexity has other consequences for the conventional analyses, and for causation.